

WORKSHEET - NAPLAN Year 9

Number and Algebra Basic Calculations and Algebra (9)

Teacher: Joanne Rust

Exam Equivalent Time: 8.75 minutes (based on NAPLAN allocation of ~1.25 min/mark)



Questions

1. ☒ Number and Algebra, NAP-66181

A butcher sells sausages in bags of four.

For a party, 20 sausages are needed.

How many bags of sausages are needed?

4 5 25 80
☐ ☐ ☐ ☐

2. ☒ Number and Algebra, NAP-95619

Cameron paid \$1.44 for 9 pencils.

How much did each pencil cost?

\$0.12 \$0.16 \$1.35 \$1.53
☐ ☐ ☐ ☐

3. ☒ Number and Algebra, NAP-13256

Ryan and Oliver are buying goldfish for their fish tank.

Oliver buys 3 times the number of goldfish as Ryan does, plus another 5.

Let m be the number of goldfish that Ryan buys.

Which expression represents the number of goldfish Oliver buys.

$(3 \times m) + 5$ $3 + (5 \times m)$ $(m \div 3) - 5$ $3 - (m \div 5)$
☐ ☐ ☐ ☐

4. ☒ Number and Algebra, NAP-60271

Brendan has 810 grams of chocolate.

He cuts the chocolate into 9 equal pieces.

How much does each piece weigh?

80 g 90 g 100 g 110 g
☐ ☐ ☐ ☐

5. ☒ Number and Algebra, NAP-07314

$$\triangle = 4$$

$$\triangle \times \text{spoon} = \text{spoon} + \text{spoon} + (3 \times \triangle)$$

What is the value of spoon ?

0 1 2 4 6
☐ ☐ ☐ ☐ ☐

6. ☒ Number and Algebra, NAP-89735

Which expression is equal to $4x - 8 + 3x + 2$?

x $7x - 6$ $7x - 10$ $x - 10$
☐ ☐ ☐ ☐

7. ☒ Number and Algebra, NAP-07312

$$8 + (17 - 5 \times 3 + 2) = \boxed{?}$$

What number does $\boxed{?}$ represent?

62 46 12 0
☐ ☐ ☐ ☐

Worked Solutions

1. Number and Algebra, NAP-66181

$$\text{Bags needed} = \frac{20}{4} = 5$$

2. Number and Algebra, NAP-95619

$$\begin{aligned}\text{Cost of 1 pencil} &= \frac{1.44}{9} \\ &= \$0.16\end{aligned}$$

3. Number and Algebra, NAP-13256

$$\begin{aligned}\text{Goldfish bought by Oliver} \\ &= (3 \times m) + 5\end{aligned}$$

4. Number and Algebra, NAP-60271

$$\begin{aligned}\text{Weight of 1 piece} &= \frac{810}{9} \\ &= 90 \text{ grams}\end{aligned}$$

5. Number and Algebra, NAP-07314

$$\begin{aligned}4 \times \text{🏸} &= \text{🏸} + \text{🏸} + 12 \\ 2 \times \text{🏸} &= 12 \\ \text{🏸} &= 6\end{aligned}$$

6. Number and Algebra, NAP-89735

$$7x - 6$$

7. Number and Algebra, NAP-07312

$$8 + (17 - 5 \times 3 + 2) = \boxed{?}$$

$$8 + 4 = \boxed{?}$$

$$12 = \boxed{?}$$